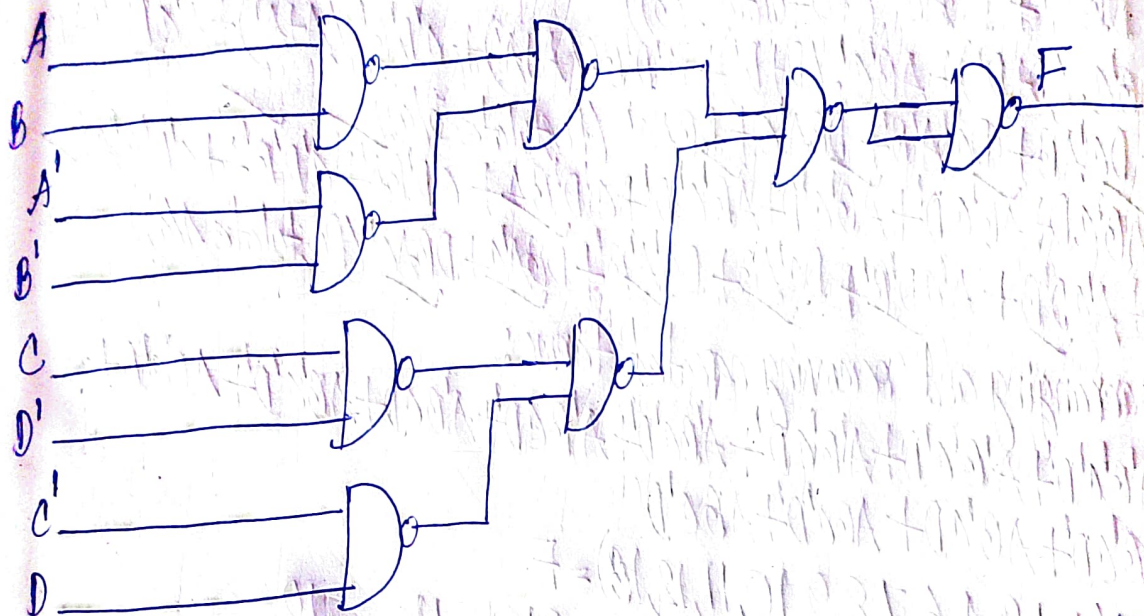
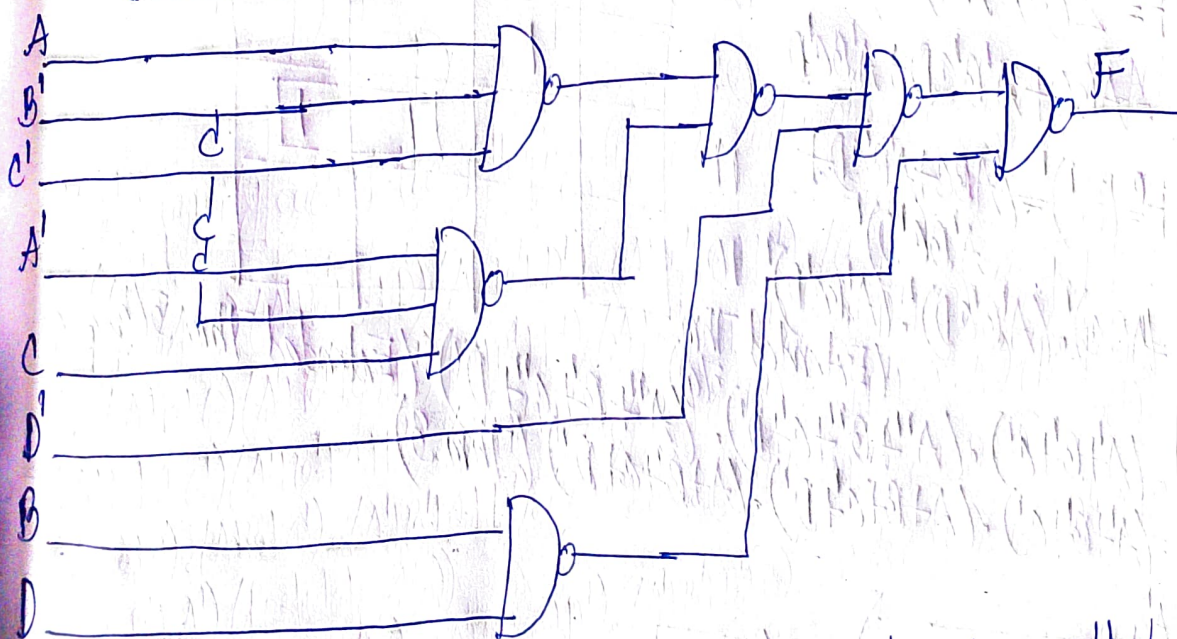


Implementation of (b)



Implementation of (a)



3-12: Implement the following functions with NOR gates. Assume that both the normal and complement inputs are available.

$$\begin{aligned}
 (a) & AB' + C'D' + A'CD' + DC(AB + A'B') + DB(AC + A'C) \\
 &= AB' + C'D' + A'CD' + DCAB + DC'A'B' + DBAC' + DBA'C \quad (\text{Postulate 4(a)}) \\
 &= AB' + C'D' + A'CD' + DCAB + DC'A'B' + DBAC' + DBA'C \quad (\text{Postulate 2(b)}) \\
 &= AB' \cdot 1 + C'D' \cdot 1 + A'CD' + DCAB + DC'A'B' + DBAC' + DBA'C \quad (\text{Postulate 5(a)}) \\
 &= AB' \cdot (C+C') + C'D' \cdot (B+B') + A'CD' + DCAB + DC'A'B' + DBAC' + DBA'C \\
 &= AB'C + AB'C' + C'D'B + C'D'B' + A'CD' + DCAB + DC'A'B' + DBAC' + DBA'C \quad (\text{Postulate 4(a)})
 \end{aligned}$$